Final Projects

ECON833, Prof. Jason DeBacker
Gyumin Kim

Abstract

This paper estimates a structural two-stage model of retail customer behavior using daily in-store data at the store–style–date level. Customers first decide whether to try on a product, and conditional on trying, decide whether to purchase it. I estimate the model using simulated maximum likelihood with a mixed logit specification for price sensitivity. The empirical results show strong negative price sensitivity, a large try-on cost, very limited information gained from trying, and meaningful age-related depreciation in style utility.

1 Introduction

Retailers frequently observe two kinds of behavioral signals: product trials (e.g., fitting room visits) and purchases. Understanding how customers transition from trying a product to buying it is central to pricing, assortment, and inventory management. My dataset contains daily style-level observations of try-on counts, sales counts, stock availability, price, and product age, aggregated at the store–style–date level.

The key question is how price and product age influence consumer decisions to try and subsequently buy. To this end, I estimate a structural two-stage discrete choice model following the framework provided in the course and subsequent faculty feedback.

2 Problem Definition

Let a customer who enters the store face a style with price p, scaled price $p_s = p/10000$, and product age A. The retailer records:

- fit_in_{sdt}: number of customers who tried the style on day t in store s,
- $sales_{sdt}$: number of purchases,
- $price_{sdt}$: posted price,
- age_{sdt}: days since launch.

The objective is to recover the underlying structural parameters governing:

1. the decision to try, and
2. the decision to purchase,

and to evaluate price sensitivity, informational content of trying, and product aging effects.

3 Model Description

Each consumer arrival (rate Λ) follows two sequential decisions.

3.1 Stage 1: Try Decision

The deterministic part of utility is:

$$\delta = v + \beta \cdot p_s + \gamma \cdot A.$$

If the consumer does *not try*:

$$U_{\text{no try}} = \log(1 + e^{\delta}).$$

If the consumer *tries*, they incur try-on cost c and learn noisy but informative fit utility:

$$U_{\text{try}} = -c + \frac{1}{\tilde{\mu}} \log\left(1 + e^{\tilde{\mu}\delta}\right).$$

The try probability is:

$$P(\text{try}) = \frac{e^{IV_{\text{try}}}}{e^{IV_{\text{try}}} + e^{IV_{\text{no try}}}}.$$

3.2 Stage 2: Purchase Decision

If the customer does not try:

$$P(\text{buy} \mid \text{no try}) = \sigma(\delta).$$

If the customer tries:

$$P(\text{buy} \mid \text{try}) = \sigma(\tilde{\mu}\delta).$$

3.3 Four Outcome Probabilities

$$\pi_{11} = P(\text{try})P(\text{buy} \mid \text{try}), \quad \pi_{10} = P(\text{try})(1 - P(\text{buy} \mid \text{try})),$$
$$\pi_{01} = (1 - P(\text{try}))P(\text{buy} \mid \text{no try}), \quad \pi_{00} = 1 - (\pi_{11} + \pi_{10} + \pi_{01}).$$

Daily expected counts:

$$E[fit\_in] = \Lambda(\pi_{11} + \pi_{10}),$$
$$E[sales] = \Lambda(\pi_{11} + \pi_{01}).$$

A mixed-logit specification is used for price sensitivity:

$$\beta_i = -\exp(\beta_{\text{mean}} + \sigma_\beta z_i), \quad z_i \sim N(0, 1).$$

4 Estimation Method

The model is estimated via Simulated Maximum Likelihood (SML):

1. Data are first cleaned, negative sales and zero-stock rows removed,
2. Aggregated at store–style–date before filtering,
3. Price scaled by 10,000,
4. A Halton sequence is used for simulation draws,
5. The likelihood uses Poisson arrival structure for try and sales counts.

Parameters estimated:

$$\theta = \{v, c, \log(\tilde{\mu}), \beta_{\text{mean}}, \log(\sigma_\beta), \gamma, \log(\Lambda)\}.$$

5 Results

Table 1 presents parameter estimates obtained from SML with 500 Halton draws.

5.1 Parameter Estimates

param	estimate	std_err	z	pval
v_intercept	-6.490118	2.156962	-3.008916	0.002622
c_trycost	63.779845	5.615066	11.358698	0.000000
log_mu_tilde	-4.610452	0.083357	-55.309405	0.000000
beta_price_mean	-14.002265	10.278671	-1.362264	0.173115
log_beta_price_sd	-3.894036	25.031060	-0.155568	0.876373
gamma_age	-0.021046	0.000846	-24.863471	0.000000
log_Lambda	0.939419	0.014924	62.945334	0.000000

Table 1: Parameter Estimates from Two-Stage Mixed Logit Model

Key findings:

- **Strong negative price effect**. The coefficient implies large sensitivity even after scaling.
- **High try cost**. Consumers face significant friction to try on products.
- **Extremely low $\tilde{\mu}$**. Trying conveys almost no new information.
- **Age depreciation**. Older products decrease in attractiveness linearly.
- **Arrival rate $\Lambda \approx 2.56$**. Each style draws roughly 2–3 daily customers.

5.2 Predicted Probabilities vs. Observed Means

Figures 1 and 2 show comparisons of predicted probabilities and observed mean outcomes across price bins.

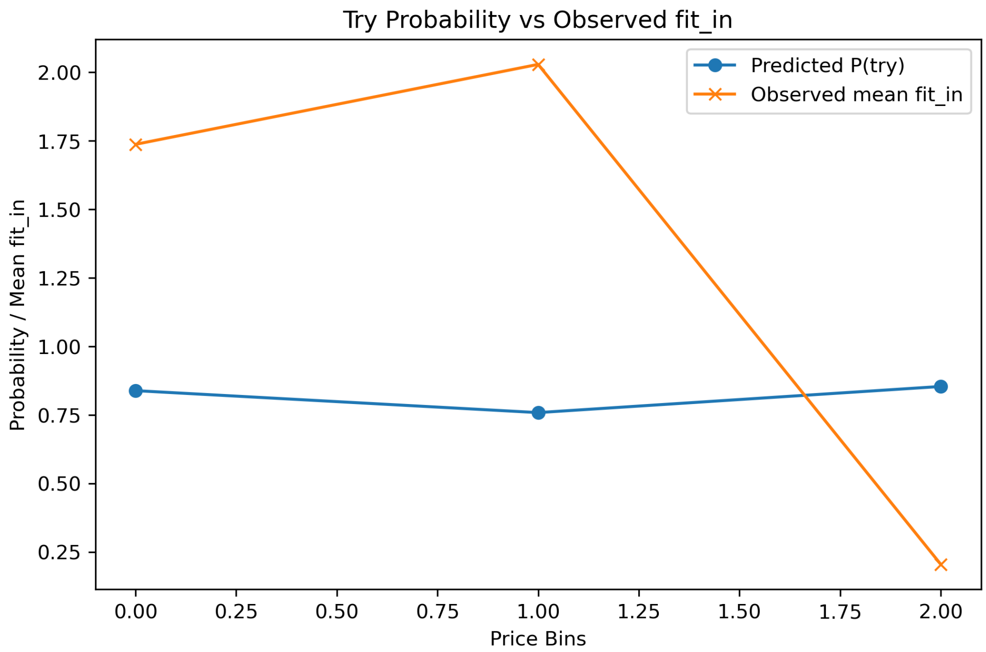

Figure 1: Predicted vs. Observed Try Probability

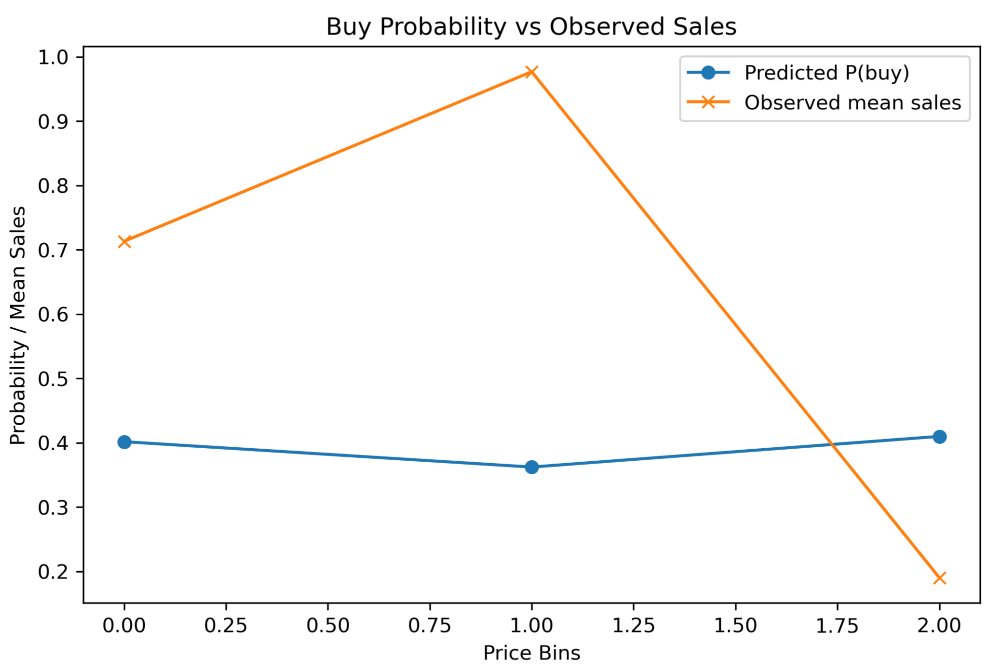

Figure 2: Predicted vs. Observed Purchase Probability

Results indicate:

- Try probability remains high (75–85%) at most prices.
- Purchase probability is 36–41%, consistent with observed conversion rates.
- Conditional purchase rate $P(\text{buy}|\text{try})$ is roughly 45–50%.

6 Conclusion

The two-stage structural model provides a sound explanation for consumer try-on and purchase behavior. Price significantly reduces utility, try-on is costly, and product age reduces attractiveness. The model matches empirical conversion patterns well and provides a foundation for richer counterfactual simulations (e.g., price optimization, style aging forecasts, or inventory-based try-on constraints).